

Report
Of
Topic

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410031318

Name of Student: PRAGYA KUMARI

Batch: 2024-2028



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Pragya Kumari

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4f7ba1e22ca32ba565fb

Student ID: STU699b4e2cd2eef1771785772



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

CANDIDATE'S DECLARATION

I PragmaKumari do hereby declare that the internship report titled “Data Science Master Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: Pragma

Student Name: PragmaKumari

Roll No.: 2410031318

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Data Science Master Virtual Internship Program. I am thankful to AICTE, the National Internship Portal, and EduSkills Foundation for facilitating this program, and to Siemens Industry Software (India) Pvt. Ltd. for supporting it. I am also grateful to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: Pragya

Student Name: Pragya Kumari

Roll No.: 2410031318

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	i
2.	Acknowledgement	ii
3.	Internship Completion Certificate	----
4.	Project Description	1
	4.1 Introduction	1
	4.2 Organization Profile	2
	4.3 Problem Statement	3
	4.4 Project Objectives	4
	4.5 Scope of the Project	5
	4.6 Technologies and Tools Used	6
	4.7 System Architecture	7
	4.8 Methodology	8
	4.9 Expected Outcomes	9
	4.10 Certificate of Completion & Communication Proof	10
5.	Bibliography / References	11

Chapter 4: Project Description

4.1 Introduction

Data Science has emerged as one of the most in-demand fields in the technology industry, combining statistics, programming, and domain knowledge to extract meaningful insights from data. With a vision to build an industry-ready workforce for emerging technologies, the All India Council for Technical Education (AICTE), through the National Internship Portal and in partnership with EduSkills Foundation, launched the Data Science Master Virtual Internship, supported by Siemens Industry Software (India) Pvt. Ltd.

I Pragya Kumari a B.Tech CSE student at the School of Computer Science and Engineering, IILM University, Greater Noida, enrolled in this 8-week virtual internship, conducted during April–June 2026. The program was designed as a self-paced, module-based learning experience covering the fundamentals of data science, statistical analysis, data visualization, and machine learning, culminating in hands-on assignments and a final assessment.

This report documents the organization behind the program, the problem addressed by the curriculum, the objectives and scope of the internship, the tools and technologies used, the methodology followed, and the outcomes achieved.

4.2 Organization Profile

The internship was jointly conducted under the following bodies:

- AICTE (All India Council for Technical Education) – A statutory body under the Ministry of Education, Government of India, responsible for planning, promotion, and regulation of technical education in the country. AICTE administers the National Internship Portal through which students across India are enrolled into industry-aligned virtual internships.
- EduSkills Foundation – A non-profit organization that partners with AICTE and leading industry players to design and deliver skill-based virtual internship programs, aiming to bridge the gap between academic curricula and industry requirements for an Industry 4.0-ready workforce.
- Siemens Industry Software (India) Pvt. Ltd. – The industry partner supporting the Data Science Master Virtual Internship, providing curriculum inputs, mentorship resources, and certification support to strengthen the practical, industry-relevant component of the program.

4.3 Problem Statement

There exists a persistent gap between the theoretical data science concepts taught in traditional academic curricula and the practical, tool-based skills expected by the industry. Students often struggle to apply statistical and machine learning concepts to real-world, messy datasets, and lack exposure to the end-to-end workflow of a data science project — from data collection and cleaning to model building and interpretation of results. The Data Science Master Virtual Internship was designed to address this gap by providing a structured, hands-on curriculum that lets students practise the complete data science pipeline in a guided, self-paced environment.

4.4 Project Objectives

The key objectives of undertaking this internship were:

- To understand the fundamental concepts of data science, including data collection, cleaning, and preprocessing.
- To learn and apply exploratory data analysis (EDA) techniques for uncovering patterns and trends in data.
- To gain proficiency in Python programming and its data science libraries for analysis and visualization.
- To understand core statistical concepts used in data-driven decision making.
- To build, train, and evaluate basic machine learning models for prediction and classification tasks.
- To develop the ability to interpret and communicate data-driven insights effectively.
- To gain exposure to an industry-recognized certification process through AICTE-EduSkills.

4.5 Scope of the Project

The scope of the internship was limited to the self-paced curriculum, assignments, and assessments prescribed by the AICTE-EduSkills Data Science Master program over the 8-week duration (April–June 2026). It covered foundational to intermediate topics in data science and machine learning suitable for undergraduate engineering students, and did not extend to advanced research-level topics such as deep learning architectures, big data infrastructure, or production-grade model deployment, which lie beyond the scope of this program.

4.6 Technologies and Tools Used

Table 4.1: Technologies and Tools Used During the Internship

Category	Tools / Technologies
Programming	Python 3
Development Environment	Jupyter Notebook / Google Colab
Data Handling	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Querying	SQL (basic querying and data extraction)
Version Control	Git and GitHub
Learning Platform	AICTE-EduSkills Virtual Internship LMS

4.7 System Architecture

A typical data science workflow, as followed during the internship modules, can be represented as a pipeline of sequential stages, illustrated below:

Data Collection	Data Cleaning	Exploratory Data Analysis	Feature Engineering	Model Building	Evaluation & Insights
----------------------------	--------------------------	--------------------------------------	--------------------------------	---------------------------	--------------------------------------

Figure 4.1: Data Science Workflow Pipeline Followed During the Internship

Each stage feeds into the next: raw data is first collected from a given source, cleaned and pre-processed to handle missing or inconsistent values, explored visually and statistically to identify patterns, transformed into meaningful features, used to train machine learning models, and finally evaluated to draw actionable insights.

4.8 Methodology

The internship followed a structured, module-based methodology broadly aligned with the CRISP-DM (Cross-Industry Standard Process for Data Mining) approach:

- Business/Problem Understanding – Studying the objective and context of each assignment given in the module.
- Data Understanding – Loading datasets and performing an initial inspection of structure, data types, and quality.
- Data Preparation – Cleaning data, handling missing values and outliers, and preparing it for analysis.
- Exploratory Data Analysis – Using statistical summaries and visualizations to identify trends and relationships.
- Modelling – Applying appropriate machine learning algorithms to the prepared dataset.
- Evaluation – Assessing model performance using relevant metrics and validating results.
- Assessment – Submitting module assignments and appearing for the final evaluation on the EduSkills LMS.

4.9 Expected Outcomes

On successful completion of the 8-week Data Science Master Virtual Internship, the following outcomes were achieved:

- A working understanding of the end-to-end data science workflow.
- Practical experience with Python and its data science ecosystem.
- The ability to perform exploratory data analysis and build basic predictive models.
- An AICTE-EduSkills certified “Data Science Master Virtual Internship” certificate, supported by Siemens Industry Software (India) Pvt. Ltd., with a final grade of O (Outstanding).
- Improved readiness for entry-level data analyst / data science roles and further academic projects in the field.

4.10 Certificate of Completion and Other Communication Proof

The official Certificate of Virtual Internship issued by AICTE, EduSkills Foundation, and Siemens Industry Software (India) Pvt. Ltd. is attached in Chapter 3 of this report (Certificate ID: 4f7ba1e22ca32ba565fb, Student ID: STU699b4e2cd2eef1771785772). [If available, attach screenshots of registration/ completion confirmation emails from the AICTE Internship Portal or EduSkills LMS here as additional proof.]

Chapter 5: Bibliography / References

- [1] AICTE National Internship Portal – <https://internship.aicte-india.org/>
- [2] EduSkills Foundation – <https://eduskillsfoundation.org/>
- [3] Siemens Industry Software – <https://www.sw.siemens.com/>
- [4] Python Software Foundation, “Python Documentation,” <https://docs.python.org/3/>
- [5] Pandas Development Team, “Pandas Documentation,” <https://pandas.pydata.org/docs/>
- [6] Scikit-learn Developers, “Scikit-learn: Machine Learning in Python,” <https://scikit-learn.org/stable/>
- [7] McKinney, W., “Python for Data Analysis,” O’Reilly Media.
- [8] James, G., Witten, D., Hastie, T., Tibshirani, R., “An Introduction to Statistical Learning,” Springer.